

Meridian Sorbent Technologies

Materials Characterisation Laboratory

Thermal screening note: composite sorbent programme

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Runs covered

Four thermogravimetric runs were made on the September 2018 samples under a single method: a 10 °C/min ramp to 500 °C in a platinum pan under nitrogen, balance purge 10 mL/min and sample purge 90 mL/min. They are serials 828 and 829 on the composite, before and after a phosphate loading cycle, and serials 830 and 831 on the two precursors.

Peak selection

The sorbents in this programme hold sorbed and structurally bound water, and every run in the set opens with a mass-loss step that is that water leaving the sample rather than the material breaking down. The bulk of that step is over by 150 °C in all four runs, though a declining tail persists to roughly 185 °C in the alginate and post-adsorption composite runs; a primary degradation peak is nonetheless read above 150 °C and the opening step is left out of the selection. The peak of record for a run is the largest rate of mass loss reached above that point, quoted with the temperature at which it occurs.

Rate basis

Mass-loss rates are read from the instrument's derivative-weight signal (Deriv. Weight, Sig6). Because the samples differ in size and in how much water they carry, rates are compared on the mass each sample still holds at 150 °C: a rate is quoted as per cent of that mass lost per °C. Peak rates in tables and traces are quoted on this basis.

Naming convention

In tables and trace plots' legends in thermal stability assessment documents, the materials should be named "Chitosan precursor", "Alginate precursor", "CTSAgn post-adsorption", and "CTSAgn as made".

Handling limits

A handling or regeneration ceiling is quoted on the material as recovered, and is the temperature at which the sample has lost five per cent of the mass it still holds at 150 °C.

Use of these runs

The runs are screening data. They characterise thermal behaviour under a nitrogen ramp and say nothing on their own about performance in service, sorption capacity or regeneration yield.